

Industrial Economics and Foreign Trade

HUT 300

Module – 2

PRODUCTION & COST

PRODUCTION

- Production is a process that create/adds value or utility
- Any economic activity which transforms inputs into outputs which are capable of satisfying human wants
- Production depends upon quantity of resources used, level of technology, size of firm, relative price of factors etc
- Inputs - Factors of production (land, labor, capital, Enterprise)

The **production function** states the technical or physical relation between the factors for production and quantity of products.

$$Q = f (L_a, L , K , E)$$

Simplest form : $Q = f(L,K)$

- **Fixed Inputs or Factors of Production:** The factors of production cannot be changed temporarily. Examples are—plants, buildings, experienced workers etc.
- **Variable Inputs or Factors of Production:** The factors of production can be changed temporarily. Some examples are—raw material, workers etc.

TYPES OF PRODUCTION FUNCTION (Based on proportion of inputs used)

❑ **Fixed Proportion Production function**

Factors of production are used in fixed proportion. Firm cannot vary the proportion of the factors, say labor and capital no matter the level of output.

❑ **Variable Proportion Production function**

Factors of production are used in variation , alternative combinations of factors can be used

❑ **Homogeneous Production function**

Type of function where scaling all inputs by the same factor scales the output by a predictable power of that factor.

$$(tx,ty)=t^kf(x,y)\text{for all }t>0$$

TYPES OF PRODUCTION FUNCTION (Based on time period)

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- **Short run production function**

Short run : Period of time in which at least one or more than one factor of production or inputs are fixed and others are variable.

Production function where at least one factor (like capital) is fixed
Eg. Restaurant can hire more staff in the short run, but needs time to expand the kitchen.

Law of variable proportion or Returns to a factor

- **Long run production function**

Long run : Period of time in which all factors of production or inputs are variable and there are no fixed factors

Production function where all factors of production can be increased or decreased.

Law of returns to scale

Concepts of Output or Product

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❖ **Total Product (TP) / Total Physical Product (TPP)/Total Returns**

Total amount of a commodity produced during given a period of time with each set of inputs

❖ **Average Product (AP)**

It is the output produced using per unit of the variable factor input.

$AP = TP / L$ (L is the variable factor in most of the cases)

❖ **Marginal Product (MP)**

Change in total product resulting from the use of one more or one less unit of variable factor.

$MP = \Delta TP / \Delta L$

Short run production function – Law of variable proportion Aparna Girish , MEC

- Examines the short run relationship between one variable input and output produced, while keeping all other factor inputs constant
- When a factor of production is increased but all other factors are fixed, then it changes the averages of factors
- Aka Law of Diminishing Returns / Law of Diminishing Marginal Product

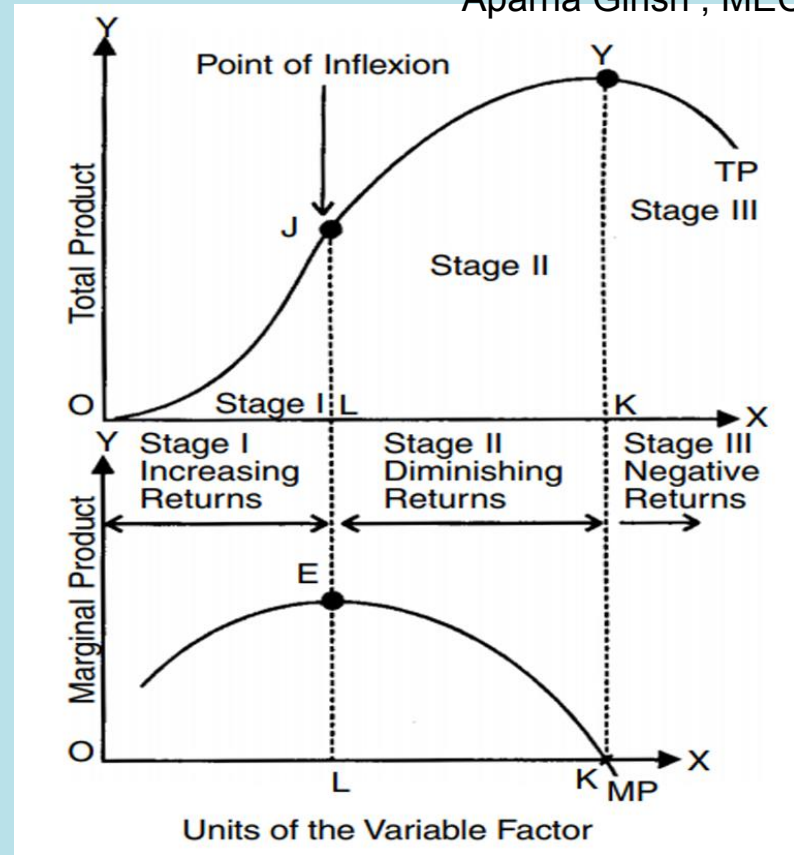
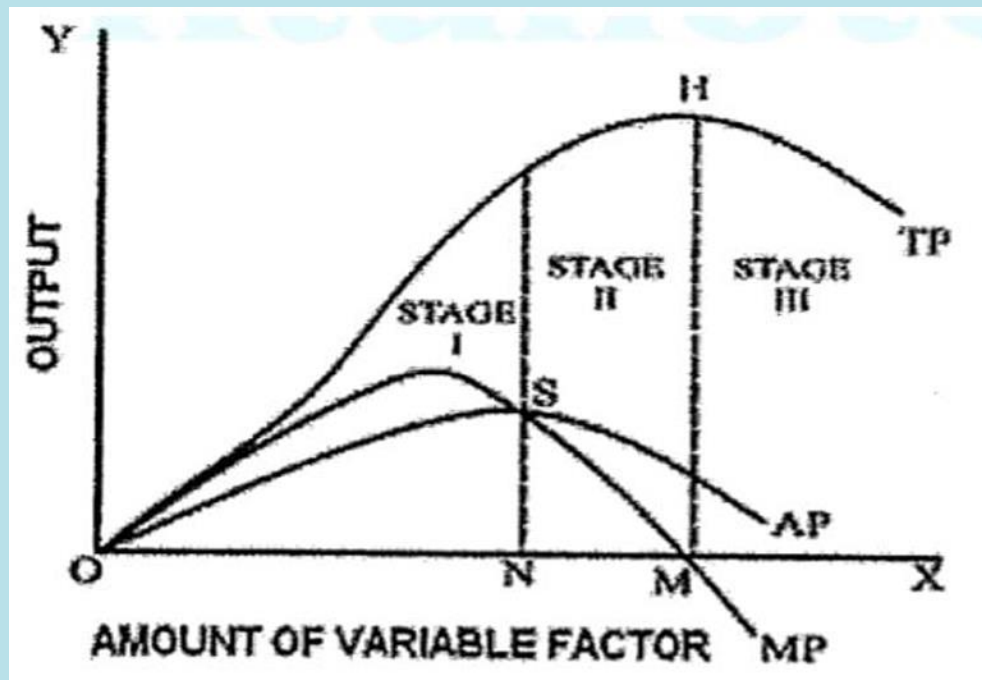
Assumptions

- One factor of production is variable while others are fixed.
- There is no change in production technique.
- The factors of production can be used in various averages. For example, one worker can be used to farm one hectare of land or 4 workers can be used to farm two hectares of land.

Statement of law

The law of variable proportion states that as more and more units of a variable factor are applied to a given quantity of a fixed factor , the total product increases at an increasing rate initially , but eventually it will increase at a diminishing rate.

Units of Labour	Total Product	Average Product	Marginal Product	Stages
1	8	8	8	Increasing Returns
2	20	10	12	
3	34	11	14	
4	46	11.5	12	
5	54	11	8	Decreasing Returns
6	56	9	2	
7	56	8	0	
8	54	7	-2	Negative Returns



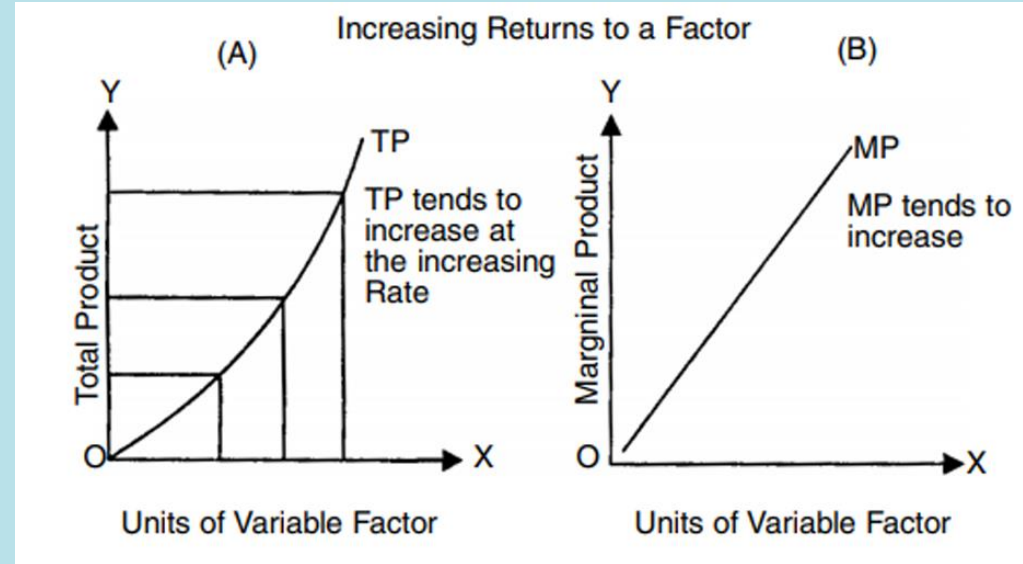
Stage 1 – Increasing returns

As the proportion of one factor in a combination of factors is increased up to a point, the marginal productivity of the factor will increase

TP , AP , MP increases at an increasing rate in the initial stage of production

Causes:

- Under-utilization of Fixed Factor
- Increase in Efficiency (division of labour and speciality)
- Better Coordination between the Factors



Stage 2 - Diminishing Returns to a Factor

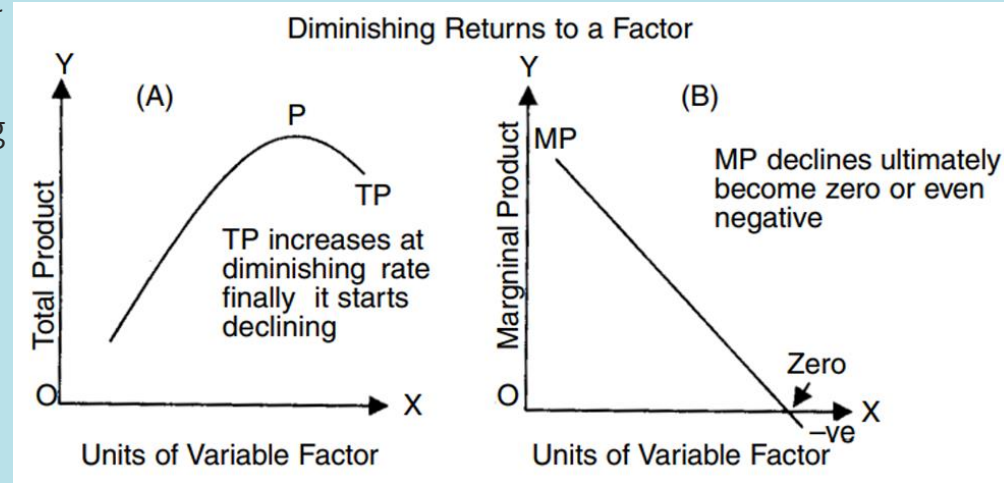
As we increase the quantity for any one input which is combined with fixed quantity of other inputs, the marginal physical productivity of the variable input must eventually decline

MP falls and TP increases at a diminishing rate

At the end of stage 2 , TP reaches maximum and MP reaches zero
AP also falls

Causes

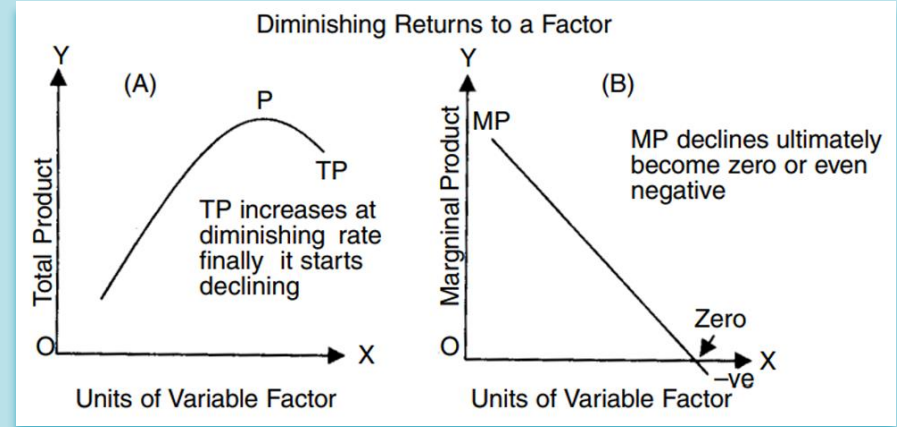
- Fixity of factors (like land)
- Imperfect substitutability of factors
- Poor coordination of factors



Stage 3 – Negative Returns to a Factor

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- MP becomes negative ,TP falls but remains positive.
- AP remains falling



Overall observations of the law of Returns to factor

When $MP > AP$, AP Rises
When $MP = AP$, AP remains constant
When $MP < AP$, AP falls

Long run production function – Law of returns to scale

- It describes the change in output when all inputs are changed in the same and equal proportion
- All factor input varies
- We have 3 stages in Returns to Scale:
 - 1.) Increasing Returns to Scale
 - 2.) Constant Returns to Scale
 - 3.) Diminishing Returns to Scale

Units of Labour (1)	Units of Capital (2)	Percentage increase in Labour and Capital (3)	Total Production (4)	Percentage increase in Total Production (5)	Returns to Scale (6)
1	2	---	10	---	Increasing
2	4	100%	30	200%	
3	6	50%	60	100%	
4	8	33%	80	33%	
5	10	25%	100	25%	Constant
6	12	20%	110	10%	
7	14	16%	120	9%	
8	16	14%	125	4%	Decreasing

Stages

Increasing Returns to Scale

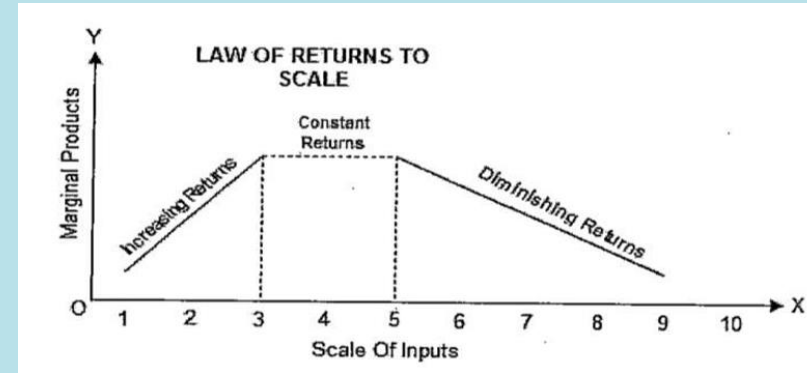
- When a given percentage increase in all factor inputs (in some constant ratio) causes proportionately greater increase in output.
- Eg . Increase in 10% of the factors labor and capital increases the production by 15%
- MP increases, TP increases at increasing rate

Constant Returns to Scale

- When a given percentage increase in all factor inputs (in some constant ratio) causes equal percentage increase in output.
- Eg. With a 10% increment of labor and capital, total returns will rise by 10%.
- MP constant, TP increases at constant rate

Diminishing Returns to Scale

- When given percentage increase in all factor inputs (in some constant ratio) causes proportionately less increase in output.
- E.g. If the total returns are increased only by 10% when factors are increased by 15%
- MP decreases, TP increases at decreasing rate



- 'Economies of scale, refer to those advantages which accrue to the firms and industry, with the expansion in the scale of production.
- Increasing the scale of production leads to a lower cost per unit of output.
- Two types

Internal Economies : It indicates those economies (or those reductions in the cost of production) which arise in the firm itself as the scale of its production increases. Like advantages enjoyed within a production unit as a result of good management, up to date machinery etc.)

External Economies : It is a situation where a firm's cost per unit of output decreases as the size of the whole industry grows. Advantages like better transport and communication linkages and trade facilities etc.

Internal Economies

- Labour Economies : Large scale production paves the way for division of labour and specialisation
- Technical Economies : More capital can help introduce up to date technologies)
- Commercial Economies : the firm places bulk orders for materials and components and enjoy lower prices for them
- Management economies : specialist management personnel can be deployed in different departments.
- Marketing Economies : In big firms the quantity bought is larger. So producer get better bargaining power.
- Financial economies : Large firms can fetch more capital as they are seen as more creditworthy
- Risk bearing economies: Large firms with diverse and multi production capability are in a better position to withstand business risks.

External Economies

Economies of concentration

Economies of information

Economies of mobilising capital

Diseconomies of scale

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Where costs per unit of output increase as the scale of production increases.

Internal diseconomies of scale: It indicates those factors which raise the cost of production of a firm as its scale of production is increased beyond a point.

- Difficult for the management to exercise control.
- The 'bulk purchase and sale' economies become diseconomies after an optimum scale. eg, advertisement expenditure and other marketing overheads will increase more than proportionately after the optimum scale.
- Financial costs will rise as growth of large firms is limited by the state of capital market

External diseconomies of scale: It refers to a situation where a firm's cost per unit of output increases as the size of the whole industry increases.

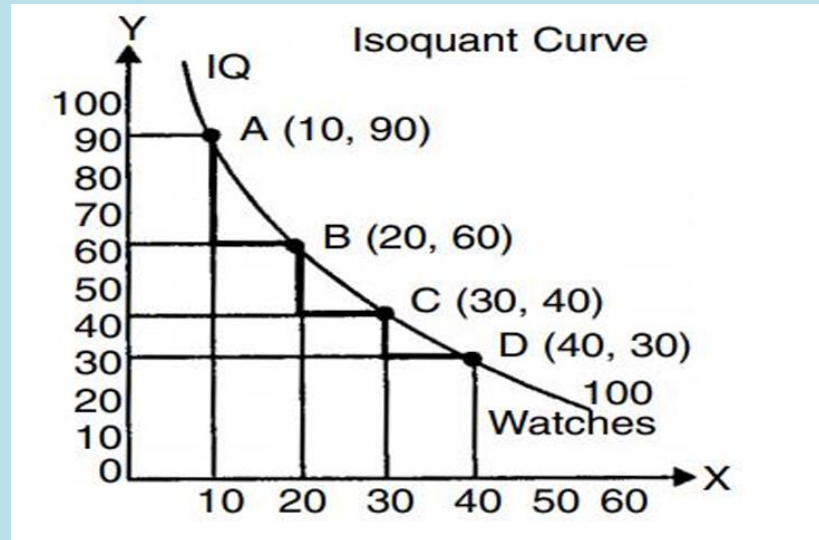
- Growing shortage of specific raw materials and skilled labour
- Disadvantage of localization of industry may appear - traffic congestion, marketing cost and housing cost, etc.
- High pollution cost

- It is a curve which shows various combinations of two factor inputs which give the same level of output.
- ISO means equal and QUANT means quantity.
- It is also called Iso-product curves and Equal product curves

Properties of Isoquants

- Downward sloping curve (because of marginal rate of technical substitution ,MRTS)
- MRTS of capital for labor may be defined as the amount of labor which can be replaced by one unit of capital given the same level of output)
- Slope of isoquant represents MRTS
- Isoquants are convex to the origin
- Two isoquants cannot cut each other
- An isoquant lying above and to the right of another isoquant represents a higher level of output

Combination	Product (Watch)	Capital (K)	Labour (L)
A	100	90	10
B	100	60	20
C	100	40	30
D	100	30	40

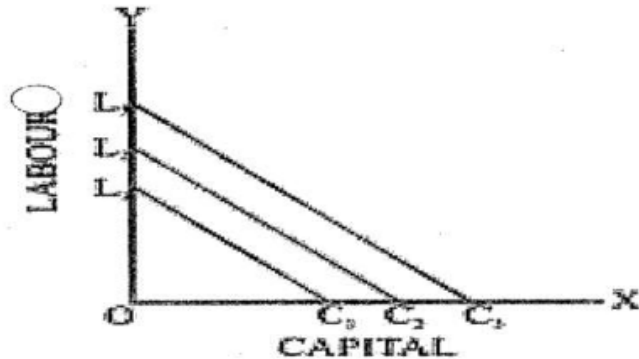


Isocost line

Total Expenditure (in ₹)	Labour $L_p = ₹ 10$	Capital $K_p = ₹ 20$
100	10	5
100	0	5
100	4	3
100	2	4

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- An isocost line is a graphical representation of various combinations of two factors (labor and capital) which the firm can afford or purchase with a given amount of money or total outlay

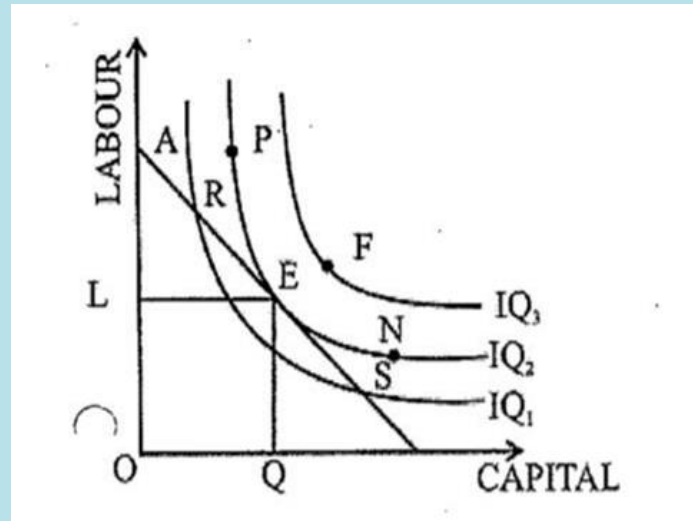


The slope of the isocost line is equal to the ratio of the prices of two factors. Thus the slope of the isocost line is given as

$$\text{Slope of isocost line} = \frac{\text{Price of Capital}}{\text{Price of Labour}}$$

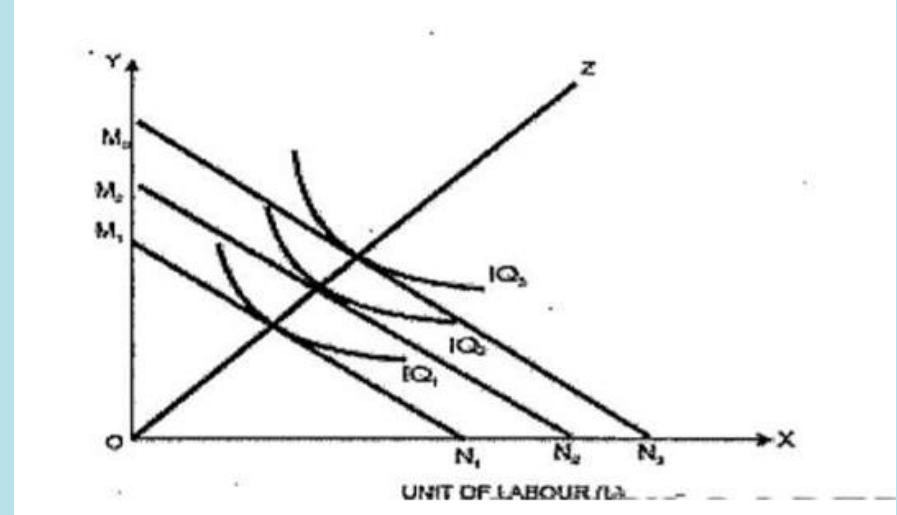
Producer's Equilibrium / Least cost combination

The point of least-cost combination of factors for a given level of output is where the isoquant curve is tangent to an isocost line.



Expansion path

- Expansion path is defined as the line joining the tangency points between various isocost lines and the corresponding highest attainable isoquants with the lowest possible isocost line.
- A line connecting all the points where different isocost lines tangent to different isoquant curves(OZ) is the expansion path of firms.

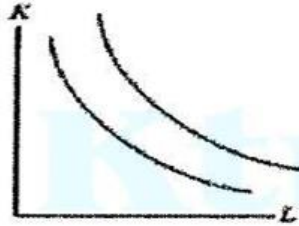


Technical Progress

- Technical progress occurs when more output is produced with same level of inputs.
- It is the result of improvements in technology and investment in research and development.
- It can be graphically represented by downward shift of isoquants since same outputs is produced with lesser quantity of inputs

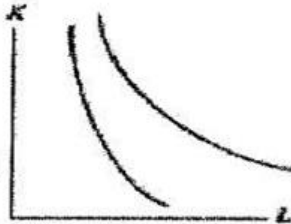
1. Neutral technical progress

It occurs when an equal reduction in both labour and capital causes a parallel downward shift of the isoquant. Here the change in the marginal product of labour and capital are the same.



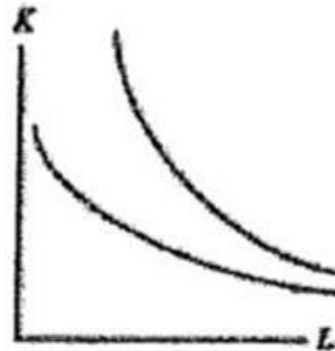
2. Labour augmenting technical progress

It occurs when as a result of a reduction in labour and capital the downward shifting isoquant becomes steeper. Here the marginal product of labour increases faster than the marginal product of capital.



3. Capital augmenting technical progress

It occurs when as a result of a reduction in labour and capital the downward shifting isoquant becomes flatter. Here the marginal product of capital increases faster than the marginal product of labour.



Cost of Production

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Cost is the expenditure incurred by a firm in the production of a commodity.

Production costs can include a variety of expenses, such as labor, raw materials, consumable manufacturing supplies, and general overhead expenses.

All of the payments or expenditures necessary to obtain the factors of production of land, labor, capital and management required to produce a commodity.

Cost Concepts

Explicit Cost: This includes those payments which are made by the producer to those factors of production which do not belong to the producer himself. Appear in the accounting records (wage, acquiring raw materials)

Implicit Cost: These costs arise in the case of those factors which are possessed and supplied by the producer himself. Cannot assign exact money value but can term them in imputed values, e.g., a producer may contribute his own building or premises for running the business, his own capital and working also as Managing Director of the firm.

Money cost: Total monetary expenses for producing a certain good. Money paid for securing factors of production.

Real Cost: Real cost signifies the aggregate of real productive resources absorbed in the production of a commodity or a service. Cost includes the pain and sacrifice involved in producing a commodity. The effort made by the owners for investment, foregoing leisure of workers etc.

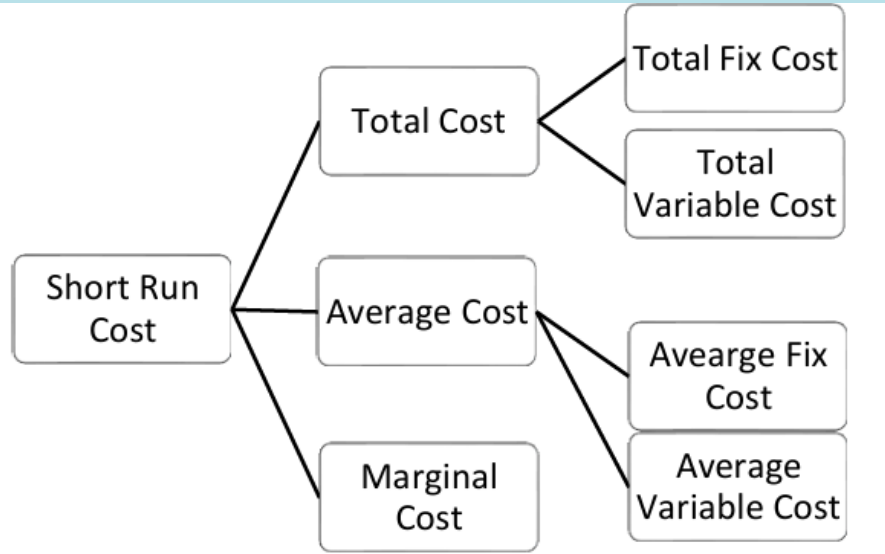
Opportunity Cost : Cost for any good is the value of the next best alternative forgone. What is forgone by using the current factors.

Private cost: It is the cost incurred by the producer/agent in the production of a good. Includes both explicit and implicit costs.

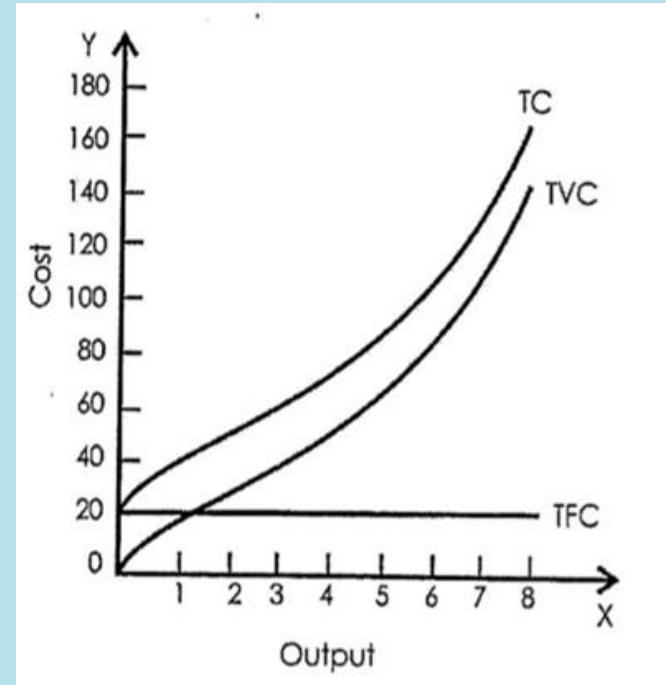
Social Cost: Total cost that the society bears on account of production. It includes private costs plus any external costs. Cost can be in the form of disutility. E.g. Pollution

Sunk cost : money that has already been spent and which cannot be recovered. It cannot be altered, increased or decreased by varying the level of activity or the rate of output. Eg. Money spent on the R&D of a product that doesn't sell well.

Concepts of cost in the Short run



*Note : It is Fixed cost ,
not fix cost*



TOTAL COST

Total expense incurred during production

The total cost is always increasing with output. The cause of this is that for increasing production always require more resources.

$$TC = TFC + TVC$$

(Here, TC = Total Cost, TFC = Total Fixed Cost, TVC = Total Variable Cost)

TOTAL FIXED COST : the cost of fixed assets.

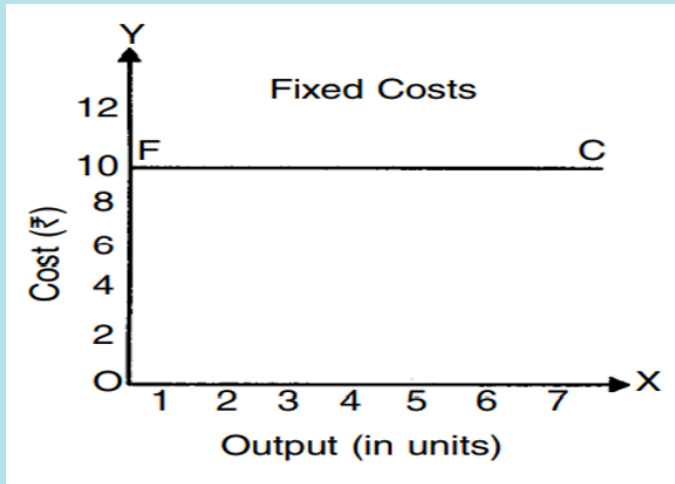
Total Fixed Cost (TFC) = Units of Fixed Resource $\times$ Price of Fixed resource

These costs do not change with the volume of production. If the output is zero, the cost will remain stable.

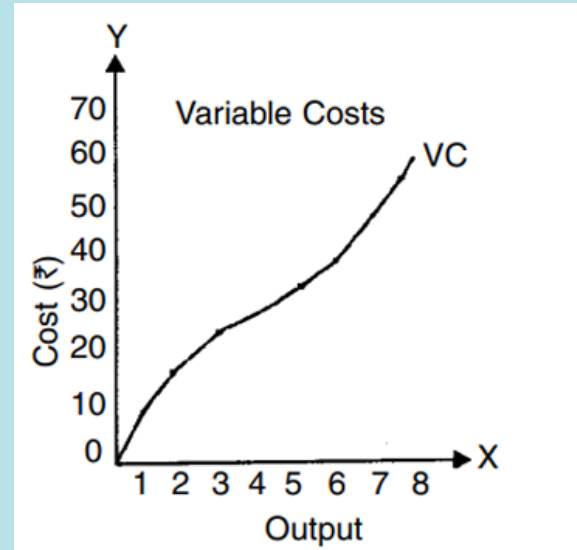
TOTAL VARIABLE COST (TVC) : sum of amounts spent for each of the variable inputs used. It varies with the level of output.

When production is zero then TVC is also zero.

Output	Fixed Cost (in ₹)
0	10
1	10
2	10
3	10
4	10
5	10
6	10
7	10
8	10



Output	Variable Cost (in ₹)
0	0
1	10
2	18
3	24
4	28
5	32
6	38
7	46
8	62



AVERAGE COST

Average cost (AC) is the cost of per unit of product.

$$AC = TC/Q$$

$$AC = AFC + AVC$$

(i) **Average Fixed Cost (AFC)** : fixed cost per unit , i.e., Total fixed cost is divided by the quantity of output

$$AFC = TFC/Q$$

(ii) **Average Variable Cost (AVC)** : Total variable cost divided by output

$$AVC = TVC/Q$$

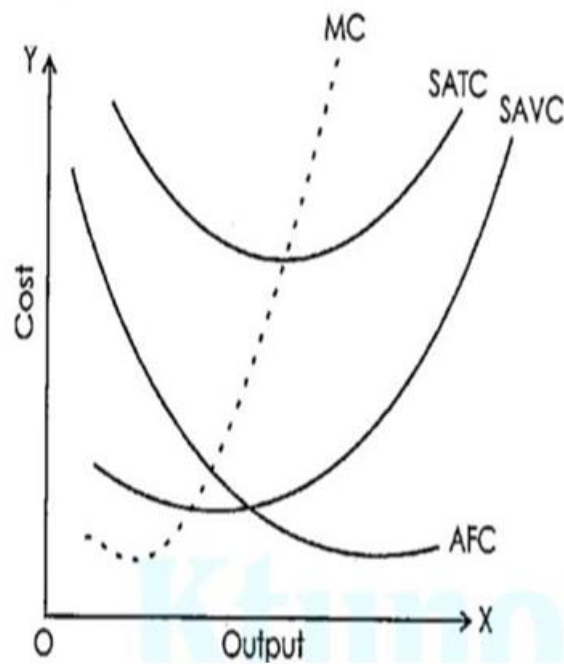
Short-term average cost curve is U-shaped.

Output	Average Fixed Cost ₹ (AFC)	Average Variable Cost ₹ (AVC)	Average Total Cost = AFC+AVC
1	10.0	10	20
2	5.0	9	14
3	3.3	8	11.3
4	2.5	7	9.5
5	2.0	6.4	8.4
6	1.7	6.3	8
7	1.4	6.6	8
8	1.2	7.8	9

Remember : Cost and Returns are related
 If your returns (output) increase quickly, then your cost per unit goes down.
 If returns increase slowly or fall, cost per unit goes up.

Reasons for the U shape of the AC curve

Diagram



The behaviour of the AC curve depends on the behaviour of the average variable cost and average fixed cost curves. The AC curve is U shaped due to:

- In the beginning both AFC and AVC curves fall. The AC curve therefore falls sharply in the beginning.
- When the AVC curve begins to rise, the AFC curve is falling steeply and the AC curve continues to fall. This is because during this stage the fall in the AFC curve is more than the rise in the AVC curve.
- But as output increases further, there is a sharp rise in AVC which more than offsets the fall in AFC. Hence AC begins to rise after a point.

Thus, the AC curve declines at first, reaches its minimum and then rises taking on a U-shape.

MARGINAL COST

Marginal cost may be defined as the additional cost of producing one more unit of output

$$MC = \Delta TC / \Delta Q = TC_n - TC_{n-1}$$

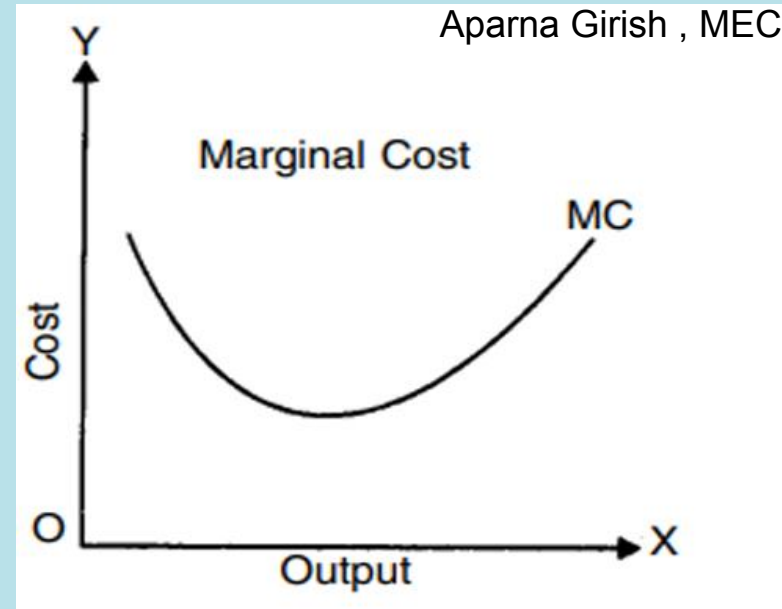
$$MC = d(TC)/d(Q) \quad (\text{First derivative})$$

(Here MC=marginal cost, TC_n = Cost of producing n th unit , TC_{n-1} = cost of producing $n-1^{st}$ unit)

ΔTC = total cost changes and ΔQ = change in volume of production

Why is MC Curve 'U' Shaped?

Law of variable proportion



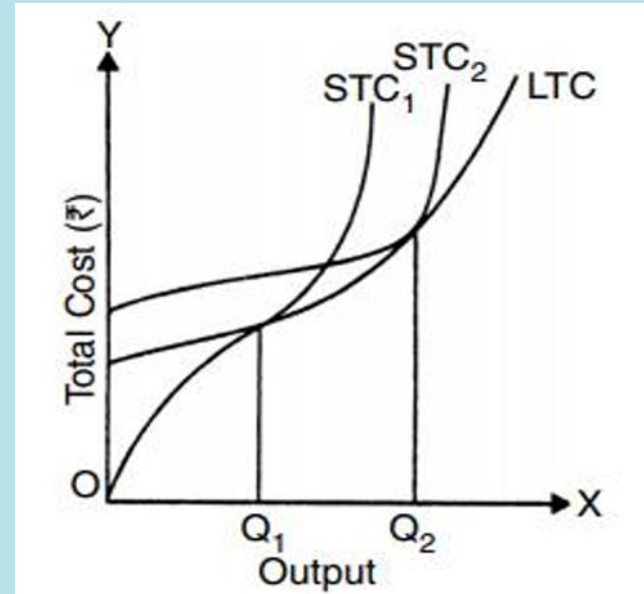
Remember : Whenever you hear a marginal concept, understand that it is the first derivative of a total concept

LONG RUN COST

The long run total cost of production (LTC) is the least possible cost of producing any given level of output when all inputs are variable.

In the long run, the output can be produced at minimum cost. This is because the firm has sufficient time in which it (i) can choose the ideal size plant (ii) Choose least Cost Factor of Production.

$$LTC \leq STC$$



Long Run Average Cost Curve or Envelope Curve

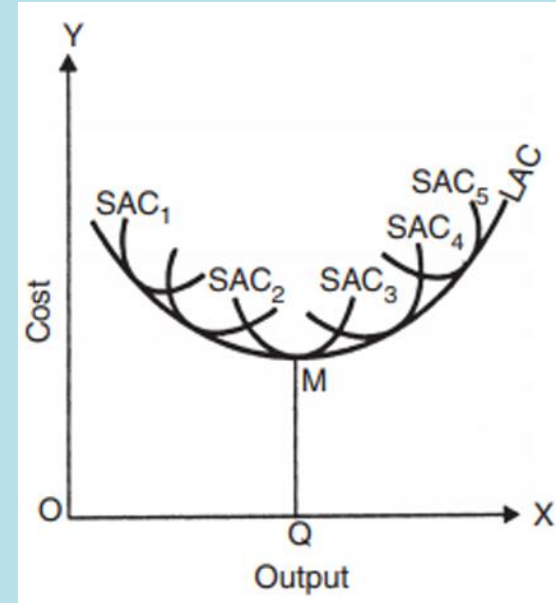
The long-run average cost (LAC) curve represents the least-cost way of producing any output level when the firm can fully adjust all inputs

In the long run, a firm can adjust/scale its plant size depending on the level of output it wants to produce.

Each plant size has its own short-run average cost (SAC) curve, which reflects the cost of production when that plant size is fixed. By combining these SAC curves for different plant sizes, we can derive the long-run average cost (LAC) curve.

The LAC curve is a smooth curve that envelops or touches the lowest points of the various SAC curves. It represents the least-cost way of producing any output level when the firm can fully adjust all inputs.

The LAC curve is also called a **planning curve**, because it helps entrepreneurs plan future expansion by choosing the most efficient plant size for any desired output.



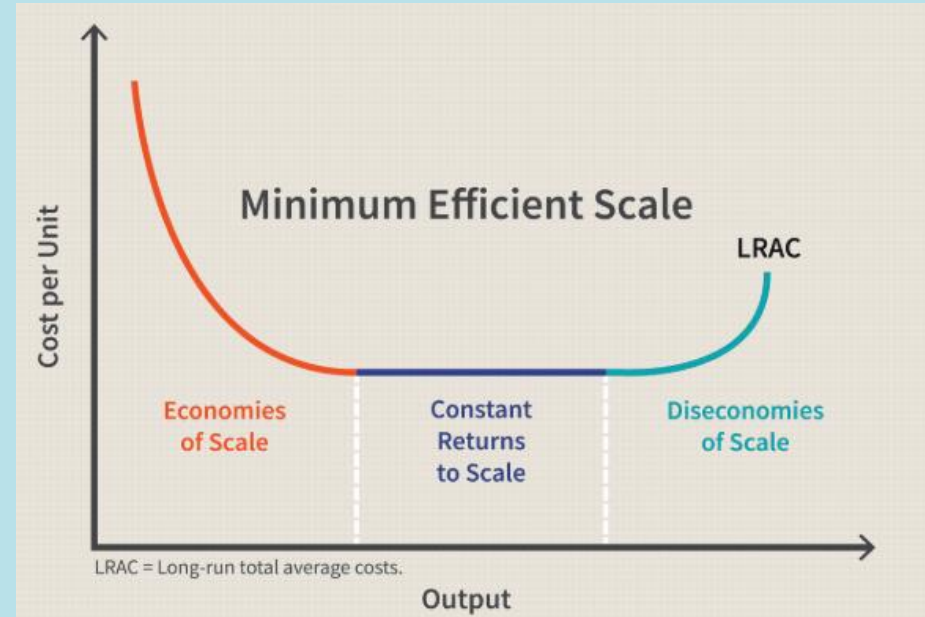
Economies of scale

How the average cost per unit changes as the scale of production increases in the long run

About **cost efficiency** as the firm expands

Economies of scale, refer to those advantages which accrue to the firms and industry, with the expansion in the scale of production

Diseconomies of scale -Where costs per unit of output increase as the scale of production increases.



Note : Returns to scale explains the technical reason for cost behavior, while economies of scale explain the cost outcome of that behavior.

Economies and Diseconomies of scale

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‘**ECONOMIES OF SCALE**, refer to those advantages which accrue to the firms and industry, with the expansion in the scale of production.

Increasing the scale of production leads to a lower cost per unit of output.

- **Internal Economies** : It indicates those economies (or those reductions in the cost of production) which arise in the firm itself as the scale of its production increases. Like advantages enjoyed within a production unit as a result of good management, up to date machinery etc.)
- **External Economies** : It is a situation where a firm's cost per unit of output decreases as the size of the whole industry grows. Advantages like better transport and communication linkages and trade facilities.

“**DISECONOMIES OF SCALE**” - Where costs per unit of output increase as the scale of production increases.

- **Internal diseconomies of scale**: It indicates those factors which raise the cost of production of a firm as its scale of production is increased beyond a point.
- **External diseconomies of scale**: It refers to a situation where a firm's cost per unit of output increases as the size of the whole industry increases.

Shut down Point- SHORT RUN

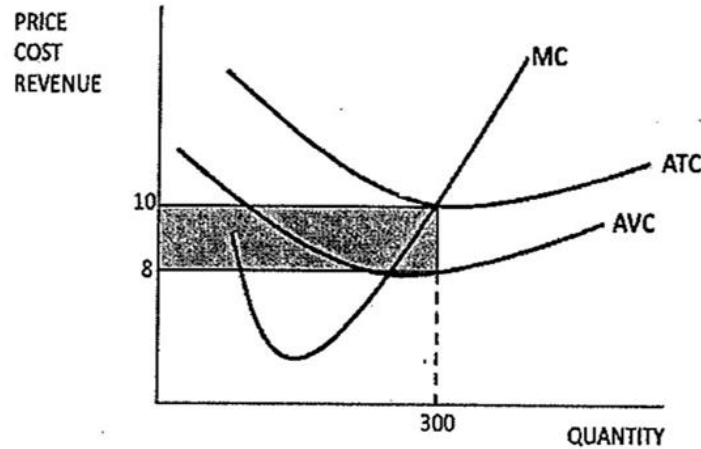
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A shut down point is a point where the firm experiences no benefit in continuing operations or production.

Point where the company earns just enough revenue to cover its total variable costs.

$$P = AVC$$

In the short run, all fixed costs have already been incurred. By shutting down a firm avoids all variable costs. However still pay fixed costs.



The shutdown price is the minimum price a business needs to justify remaining in the market in the short run. In the figure, firm is at its shut down point when the price of its product becomes Rs.8. The shaded area represents the fixed cost of the firm. At this price, the firm is not covering any of its fixed cost. At any price between Rs.8 and Rs.10, it will be able to pay off some of its fixed costs, so it makes sense to keep operating.

Revenue

It is the total amount of money received by a firm from the sale of goods and services during a certain period

Total Revenue (TR) - Total revenue is equal to the number of units of output sold multiplied by the price per unit.

$$TR = P \times Q$$

Average Revenue (AR) -revenue per unit of commodity sold.

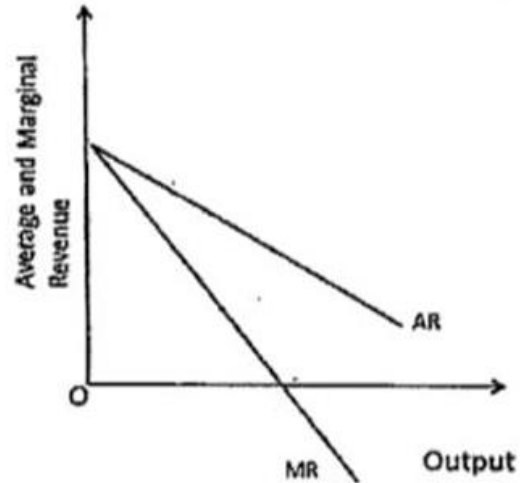
$$AR = TR/Q$$

Marginal Revenue (MR)

It is the addition to total revenue from the sale of an additional unit of commodity.

$$MR = TR_n - TR_{n-1}$$

$$MR = d(TR)/dQ$$



Both AR and MR decline as a firm increases its output. This is because a firm can sell large quantities only at lower prices. In that case, the average revenue (price) of the product falls. When AR falls MR will also fall. But fall in MR will be more than the fall in the AR. Hence the marginal revenue curve will always lie below the average revenue.

If the average revenue (price) remains constant, the marginal revenue will also remain constant and will coincide with the average revenue.

BREAK EVEN ANALYSIS

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Breakeven analysis studies the relationship between total costs, total revenue and total profit / losses over the whole range of output

A breakeven situation (zero profit) occurs whenever total revenue equals total cost.

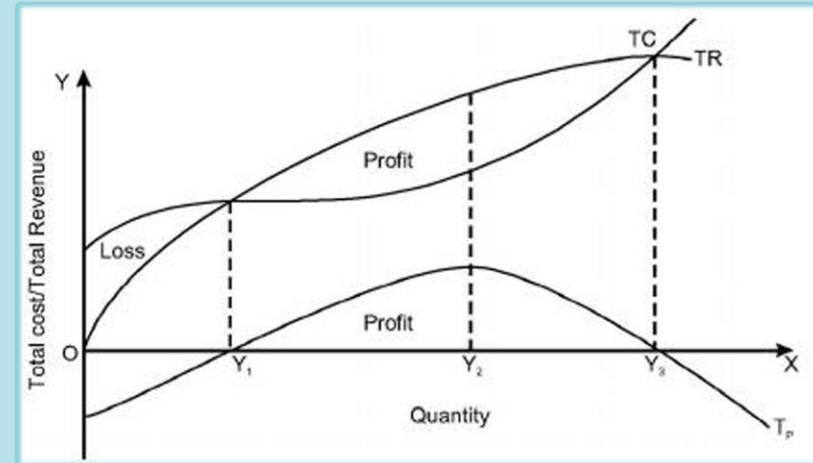
Breakeven condition occurs at two different output levels - Y_1 and Y_3 .

Below Y_1 , losses will be incurred because $TR < TC$.

Between Y_1 and Y_3 , profit will be obtained because $TR > TC$.

Above Y_3 , losses will occur again because $TR < TC$.

Total profit is maximized within the range of Y_1 to Y_3 , where the vertical distance between the TR and TC curves is greatest, that is at an output level of Y_2



Non- linear case

TR is a straight line assuming that firms charge a constant selling price P per unit of output.

Breakeven point is Y_b

Total revenue, $TR = p \times Q$

Total cost, $TC = F + v \times Q$

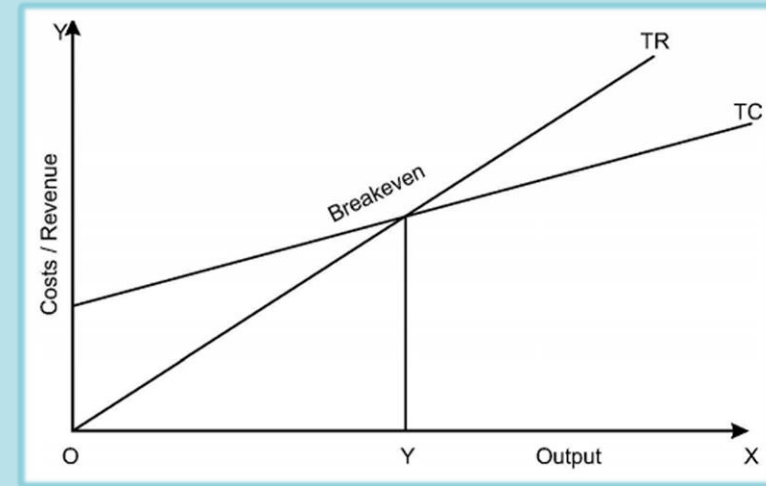
(p = selling price per unit, F = fixed cost, v = variable cost per unit , Q = volume of output)

Now break-even output level is that level where profit is zero.

$TR = TC$.

$p \times Q = F + v \times Q$

Breakeven point in units : $Q = F / (p - v)$ units



Linear case